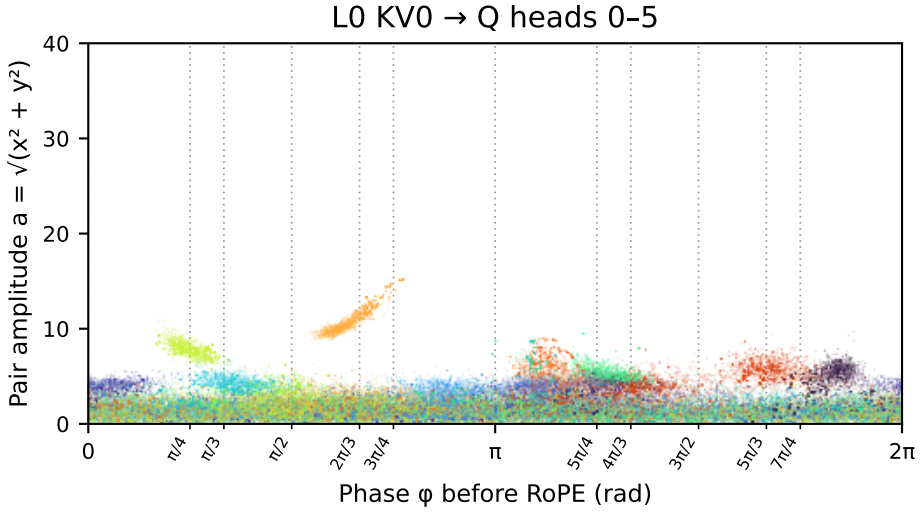
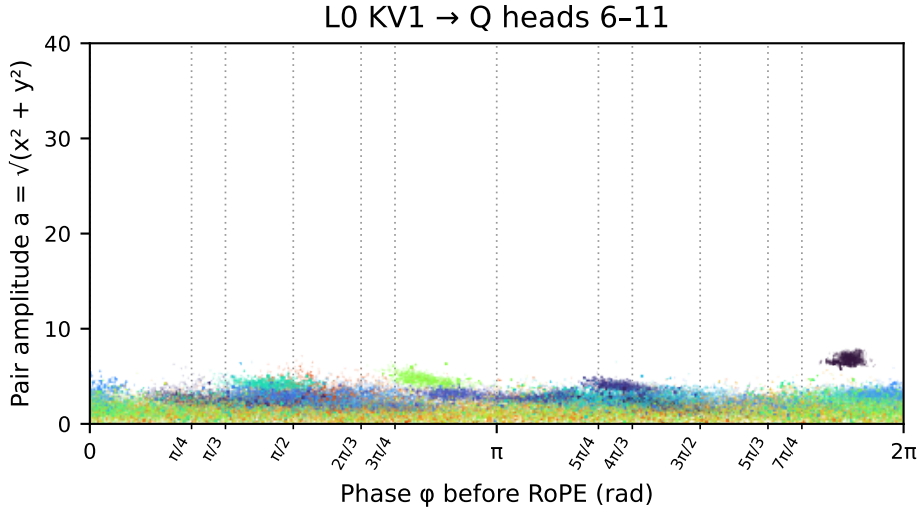
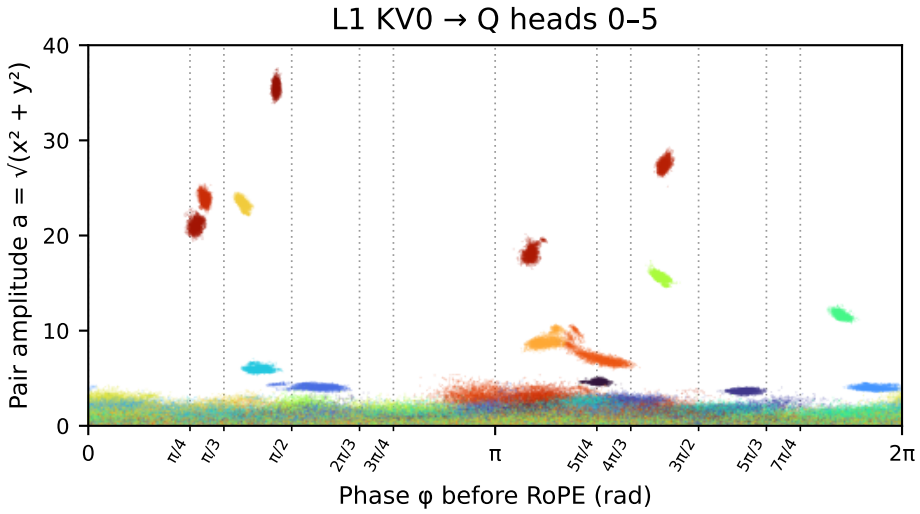
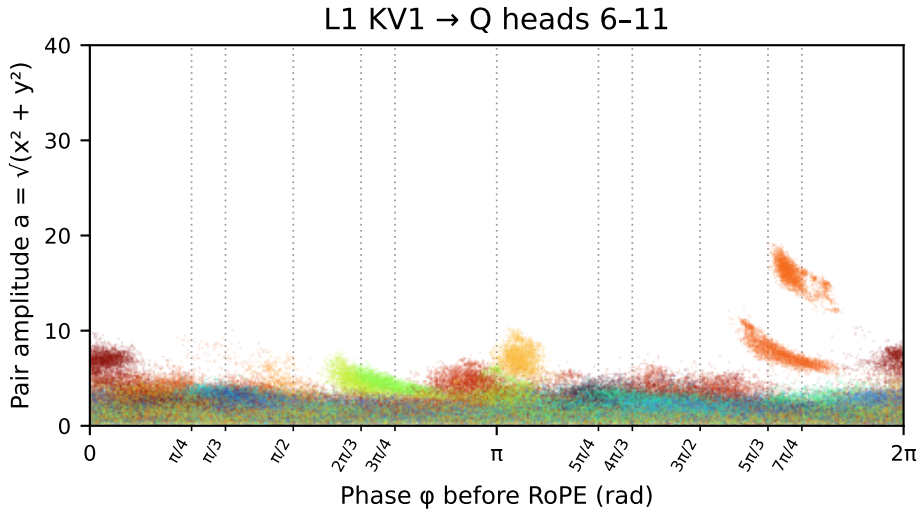
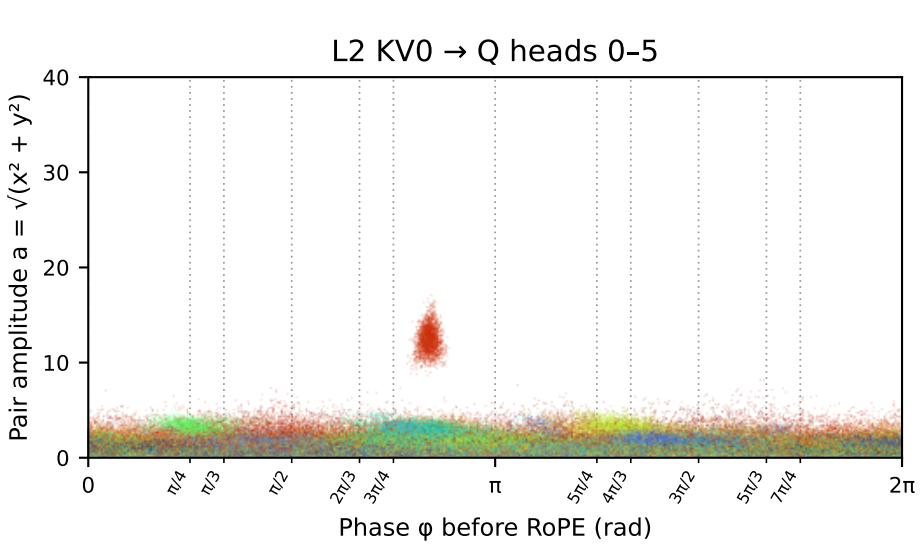
Qwen keys · layer 0–3 · amplitude axis 0–40

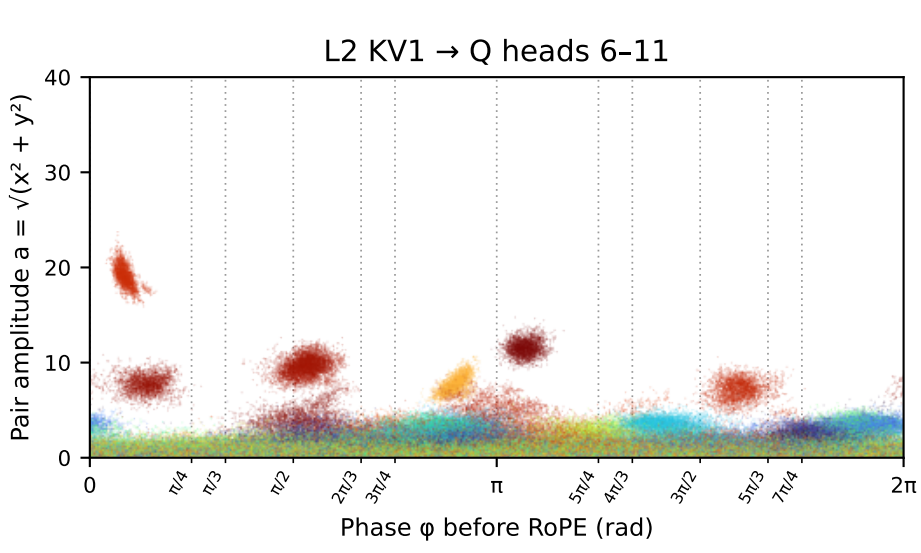


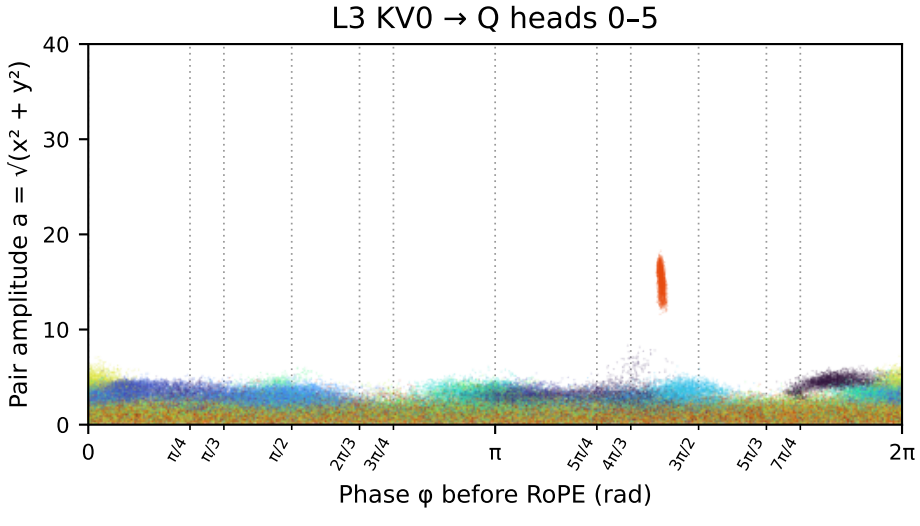


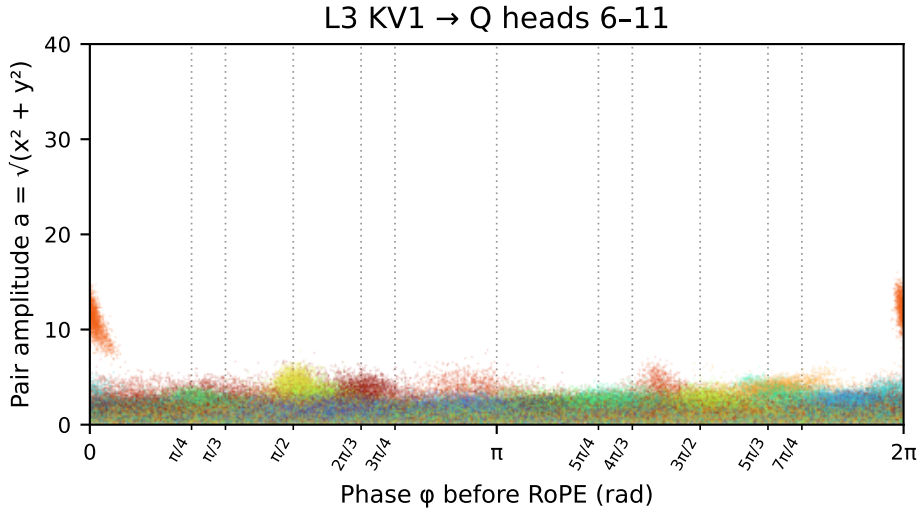


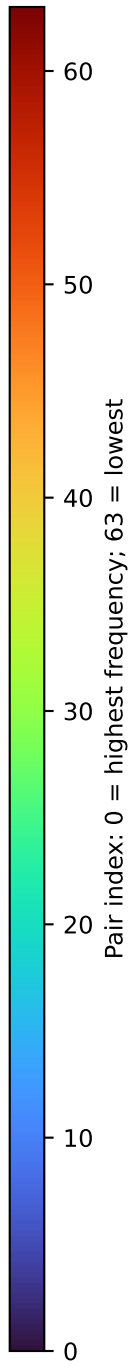






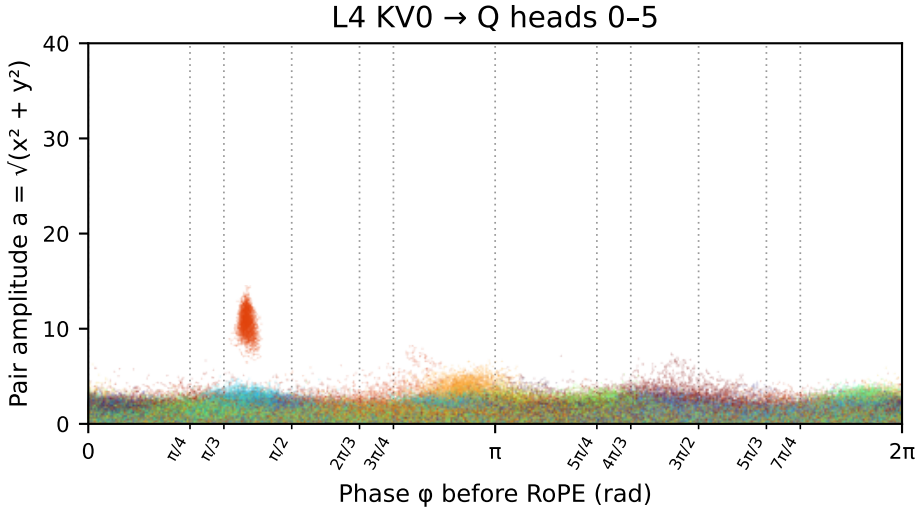


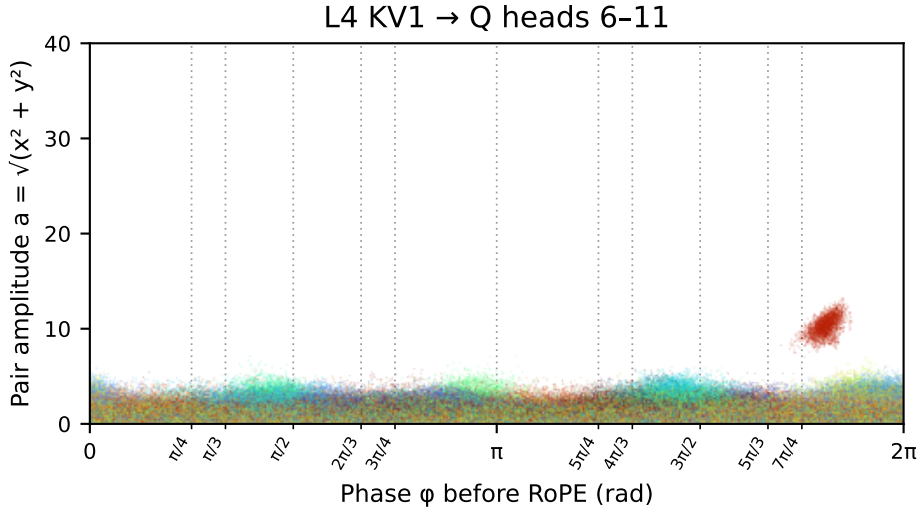


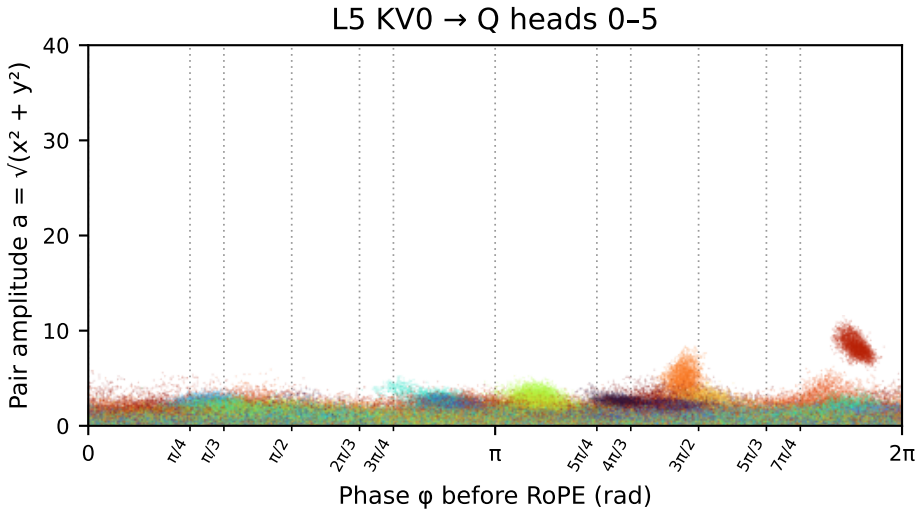


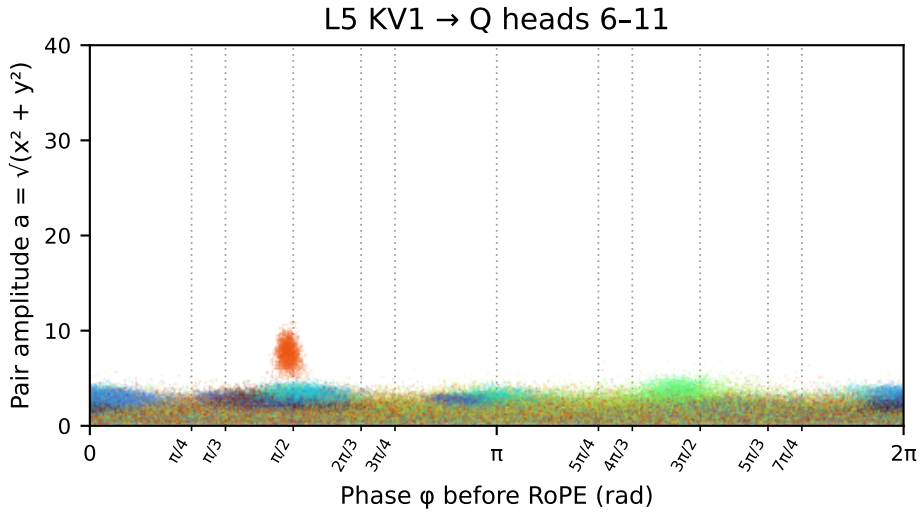
Every point = one token × one RoPE pair. 2,379 prose key tokens (chat-prefix keys excluded), 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Two shared key heads per layer: KV0 serves Q heads 0–5; KV1 serves Q heads 6–11.
Gray dotted guides mark multiples of π/4 and π/3.

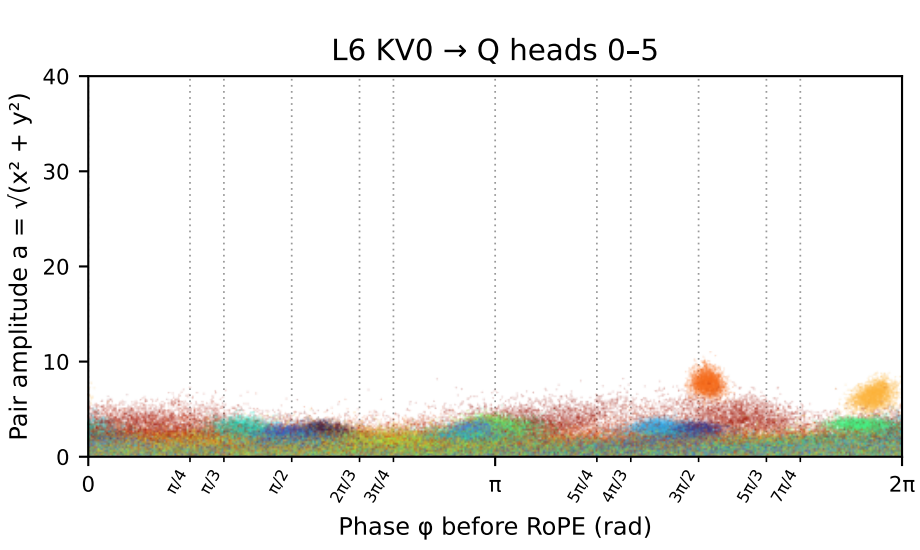
Qwen keys · layer 4–7 · amplitude axis 0–40

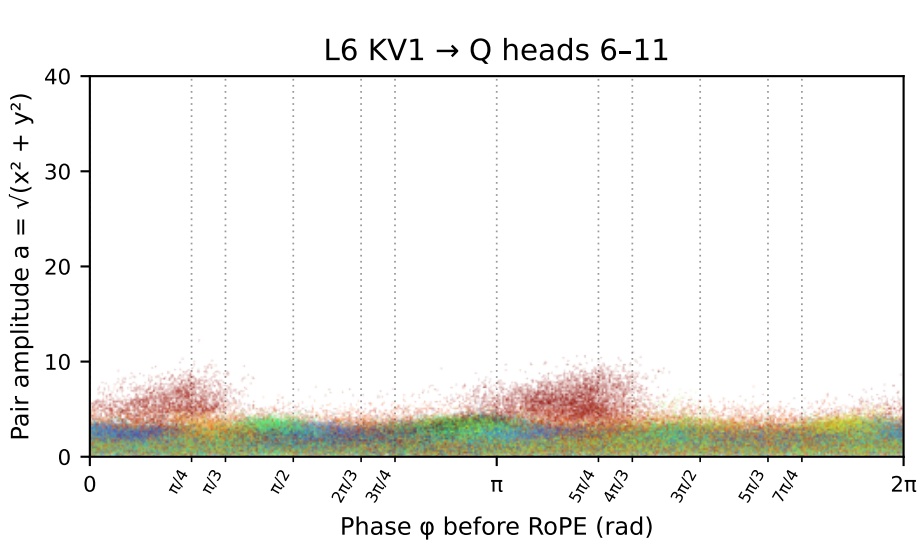


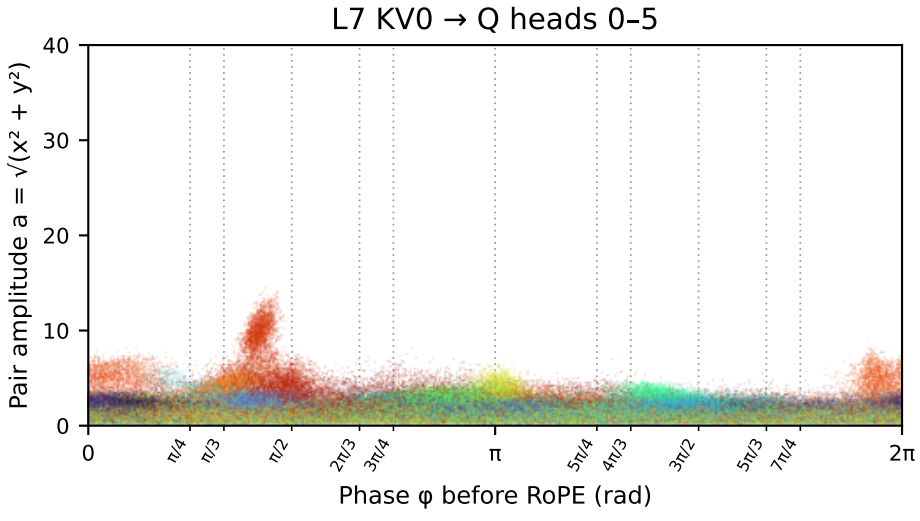


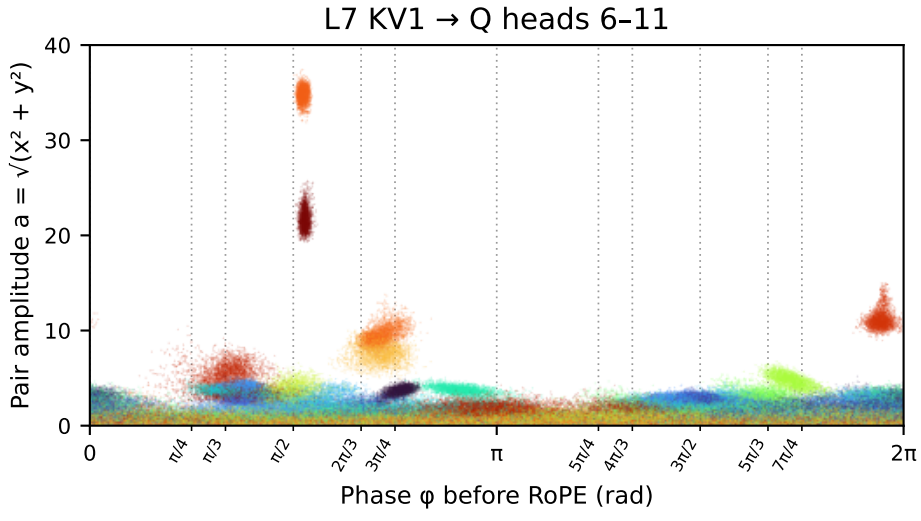


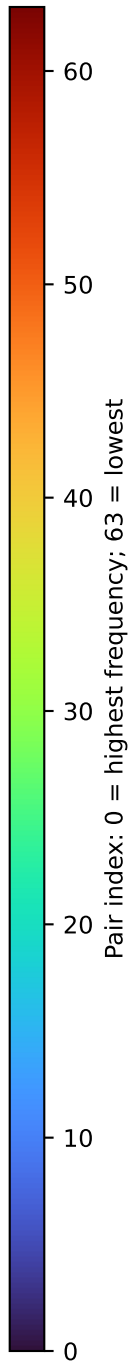






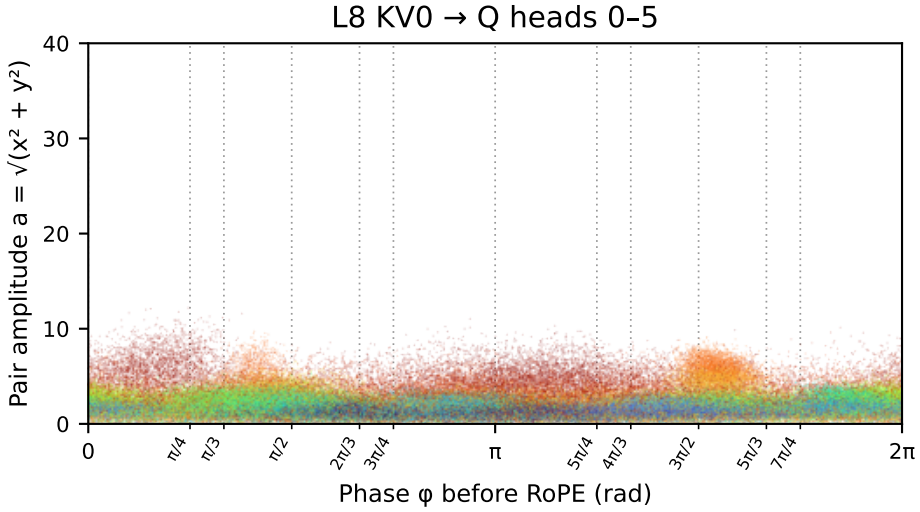


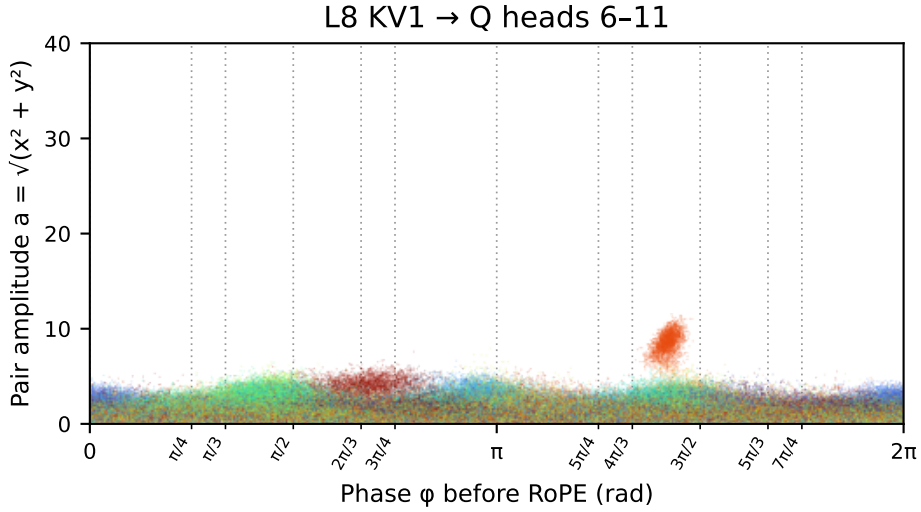


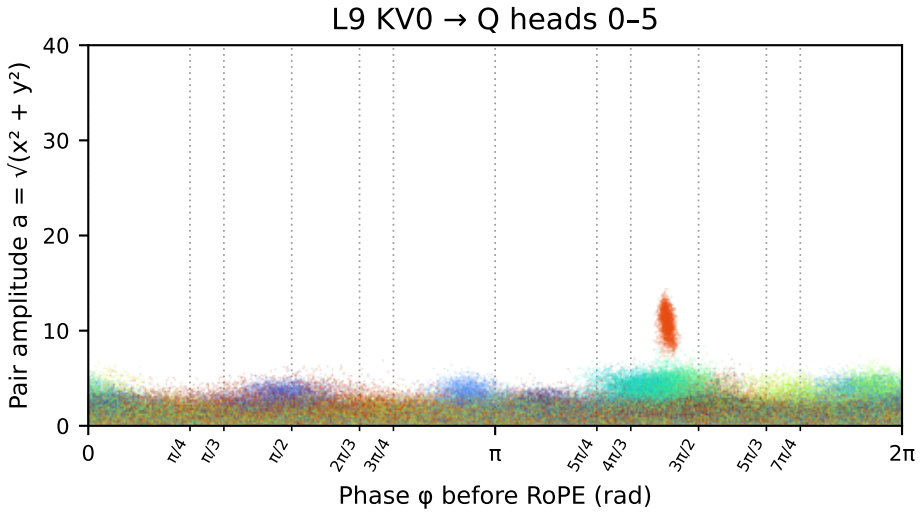


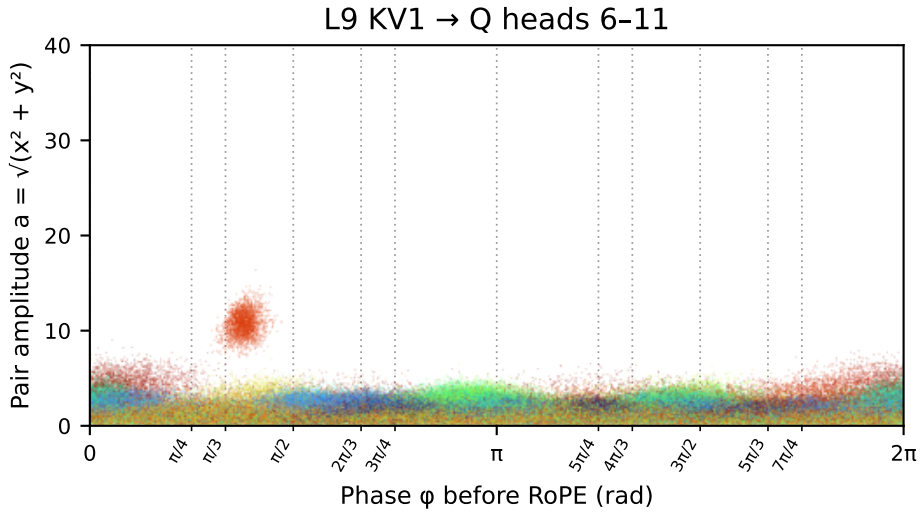
Every point = one token × one RoPE pair. 2,379 prose key tokens (chat-prefix keys excluded), 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Two shared key heads per layer: KV0 serves Q heads 0–5; KV1 serves Q heads 6–11.
Gray dotted guides mark multiples of π/4 and π/3.

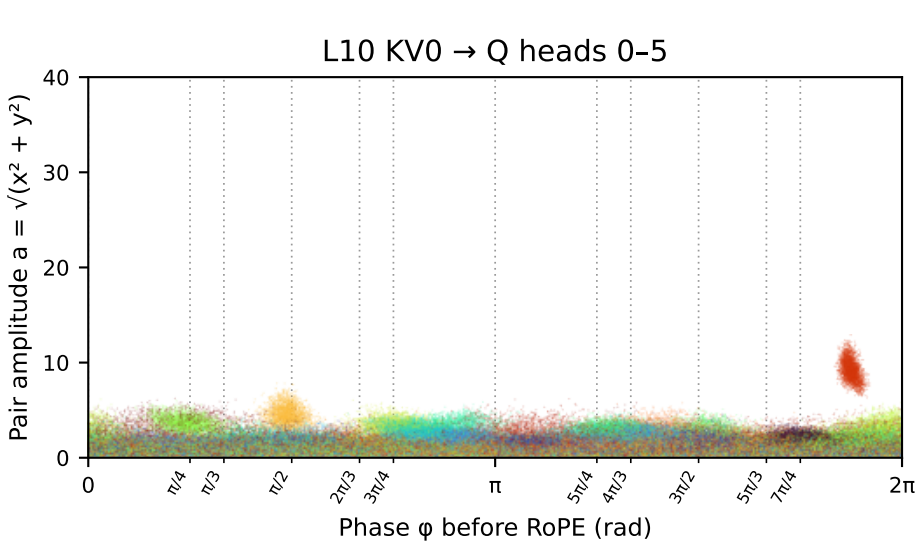
Qwen keys · layer 8–11 · amplitude axis 0–40

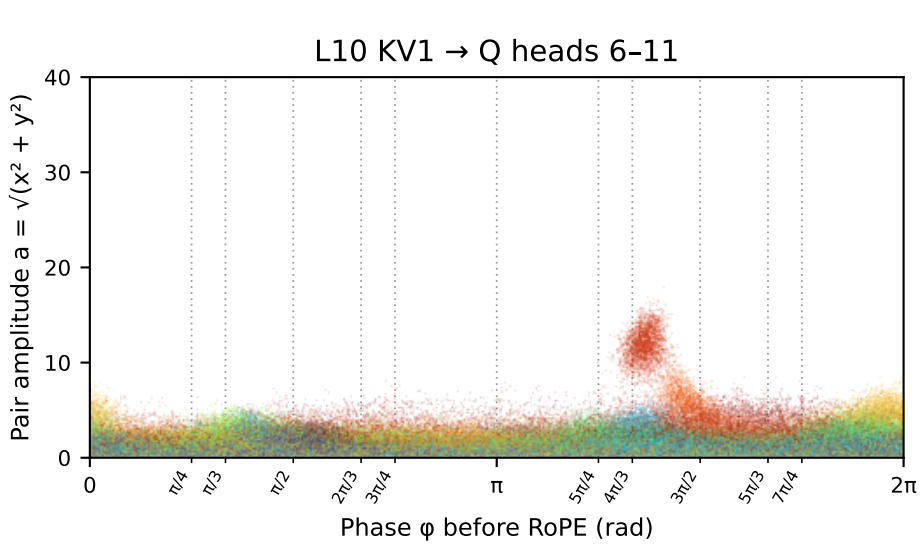


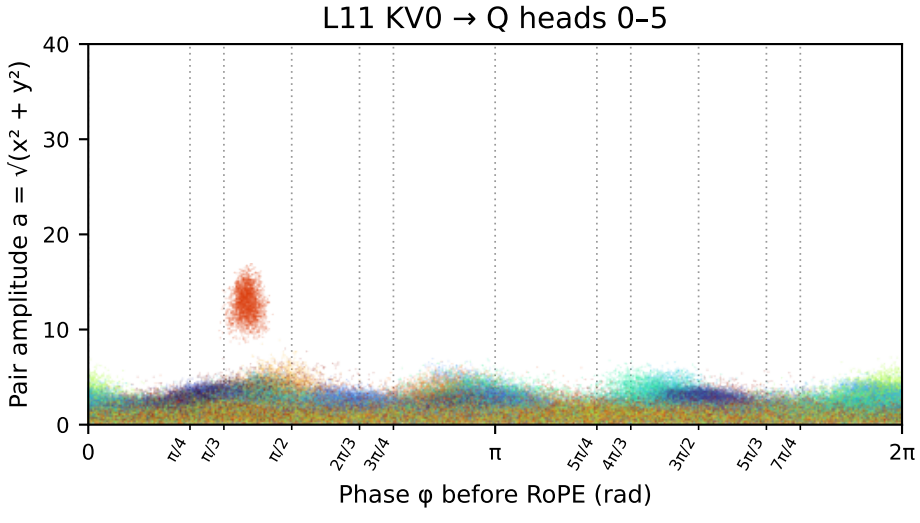


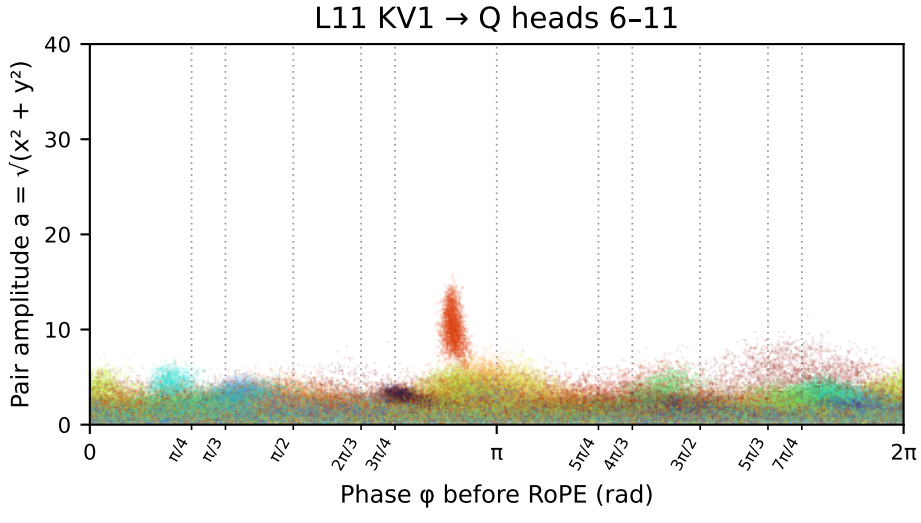


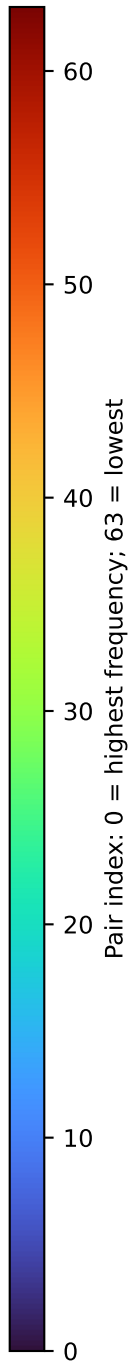






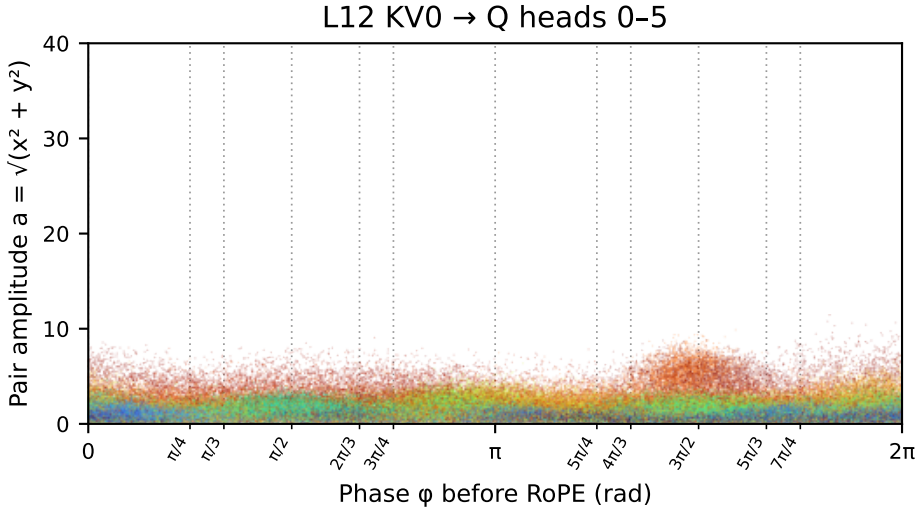


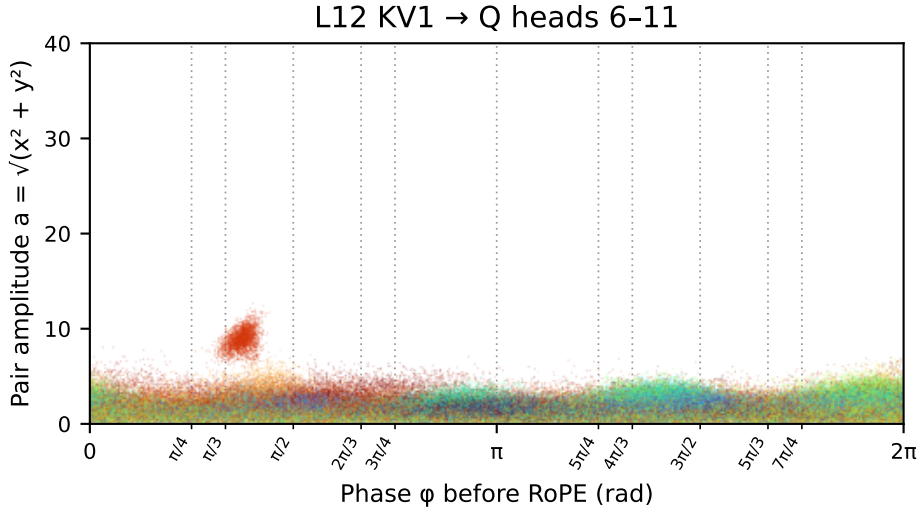


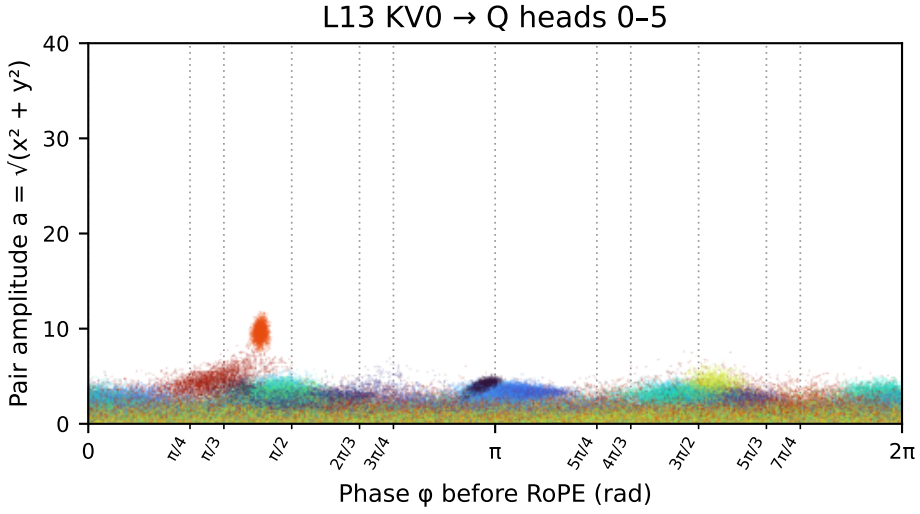


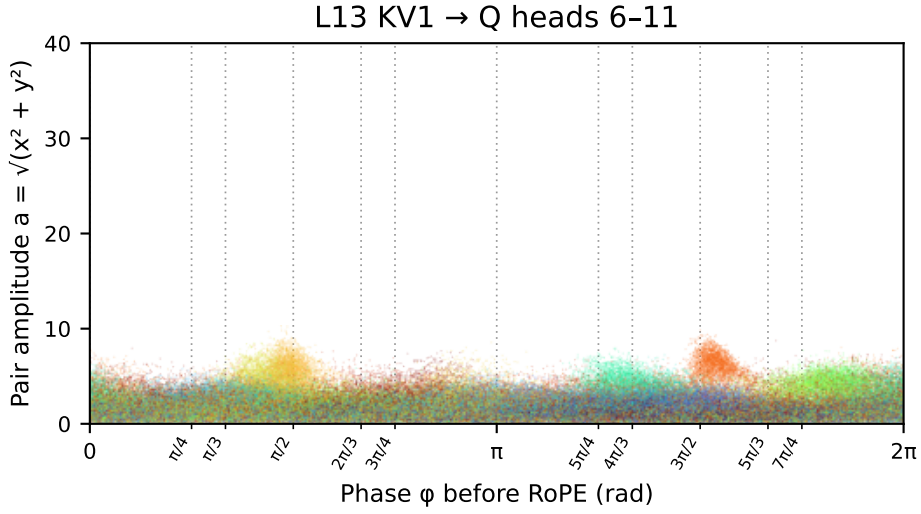
Every point = one token × one RoPE pair. 2,379 prose key tokens (chat-prefix keys excluded), 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Two shared key heads per layer: KV0 serves Q heads 0–5; KV1 serves Q heads 6–11.
Gray dotted guides mark multiples of π/4 and π/3.

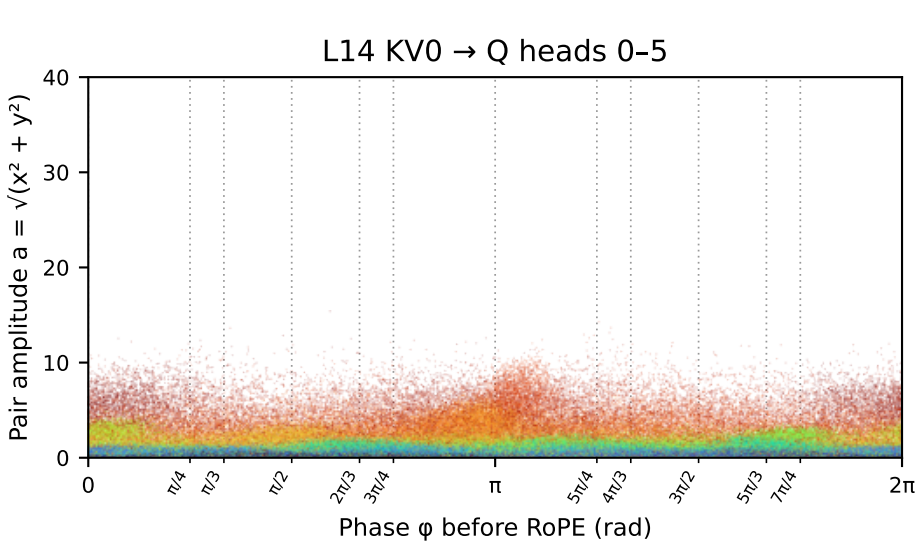
Qwen keys · layer 12–15 · amplitude axis 0–40

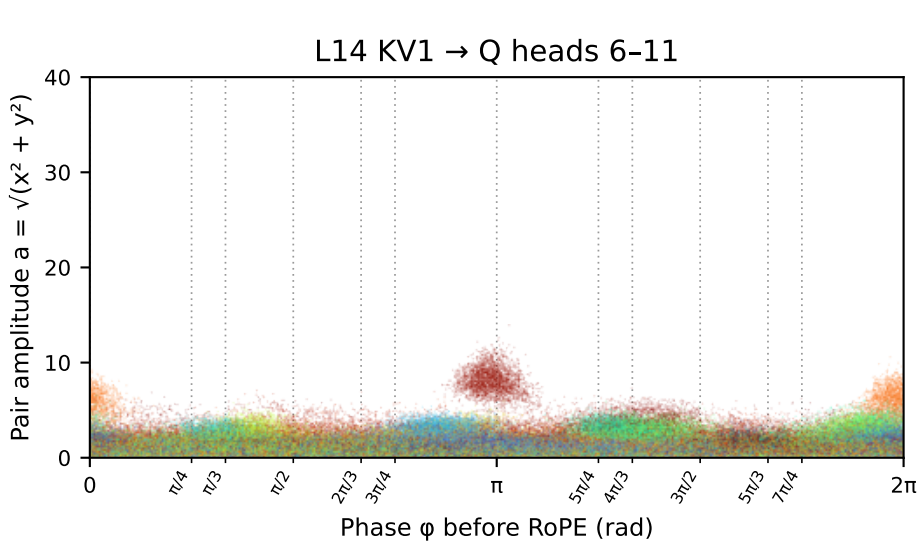


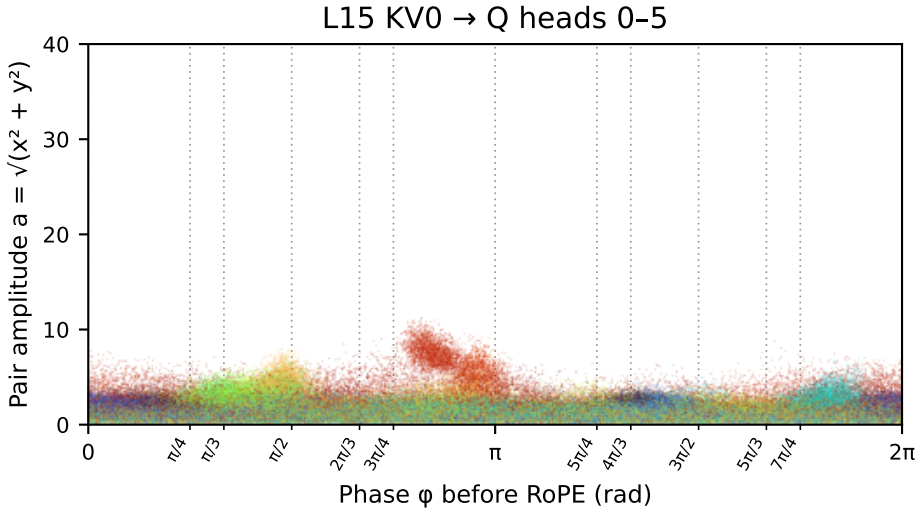


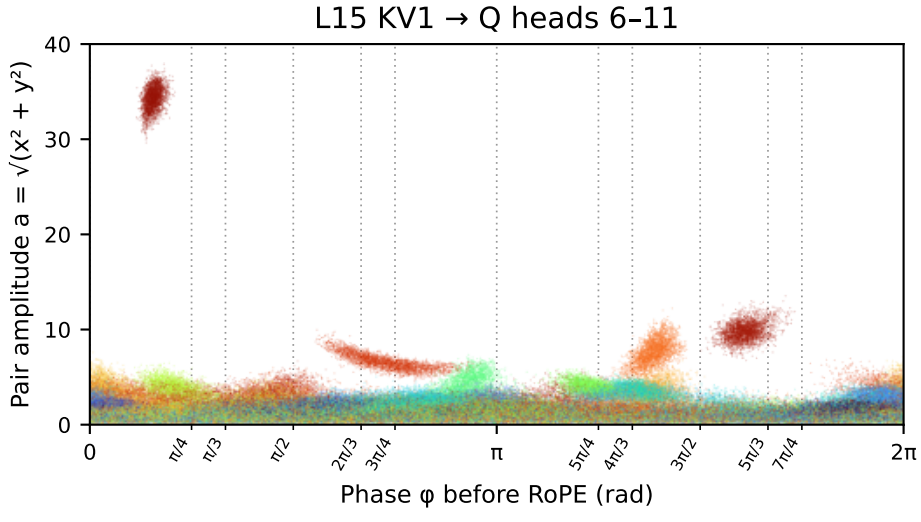


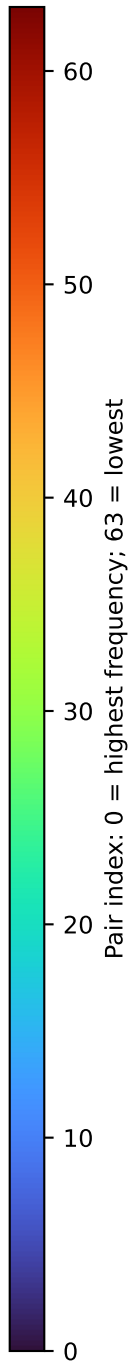






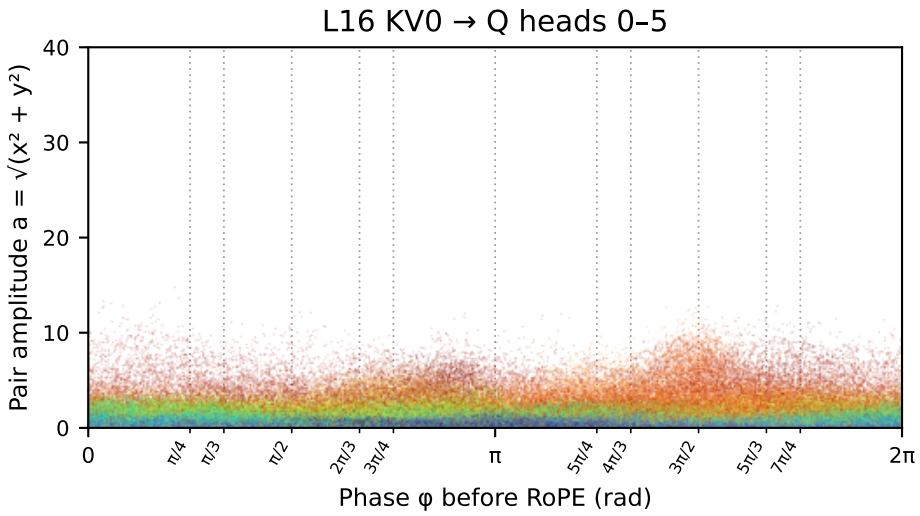


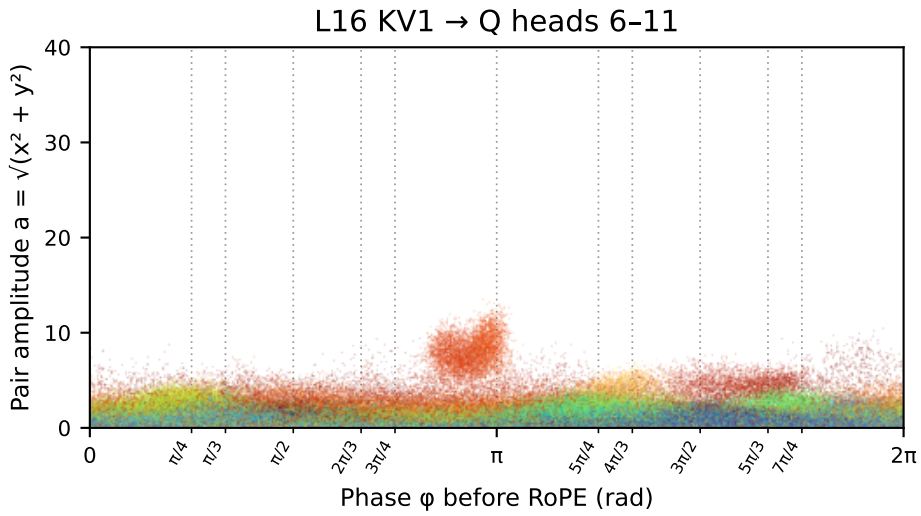


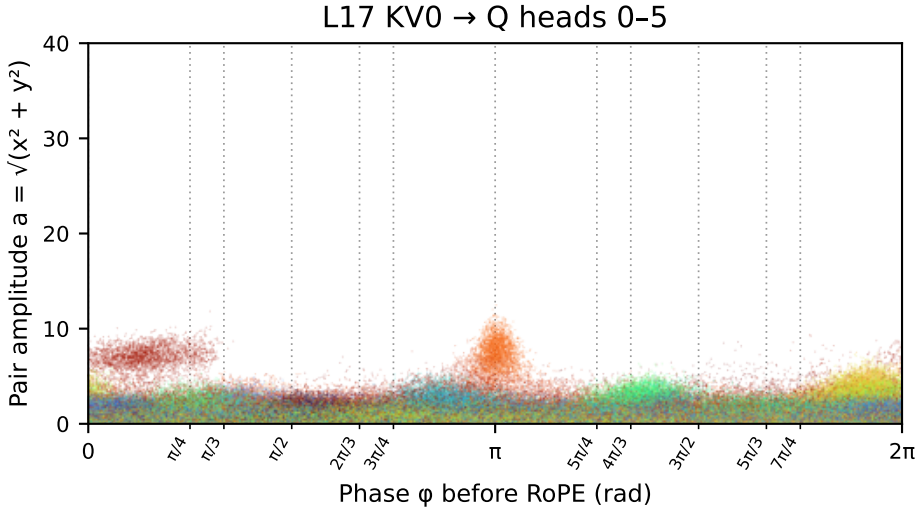


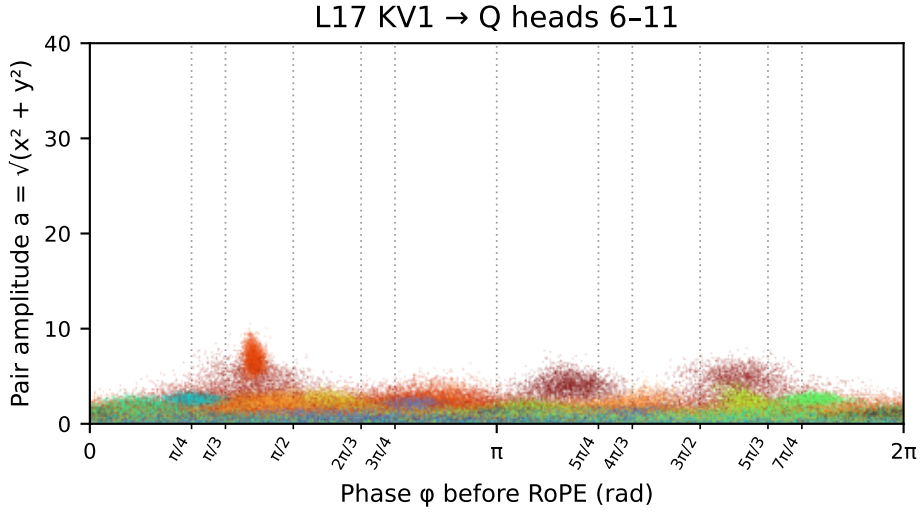
Every point = one token × one RoPE pair. 2,379 prose key tokens (chat-prefix keys excluded), 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Two shared key heads per layer: KV0 serves Q heads 0–5; KV1 serves Q heads 6–11.
Gray dotted guides mark multiples of π/4 and π/3.

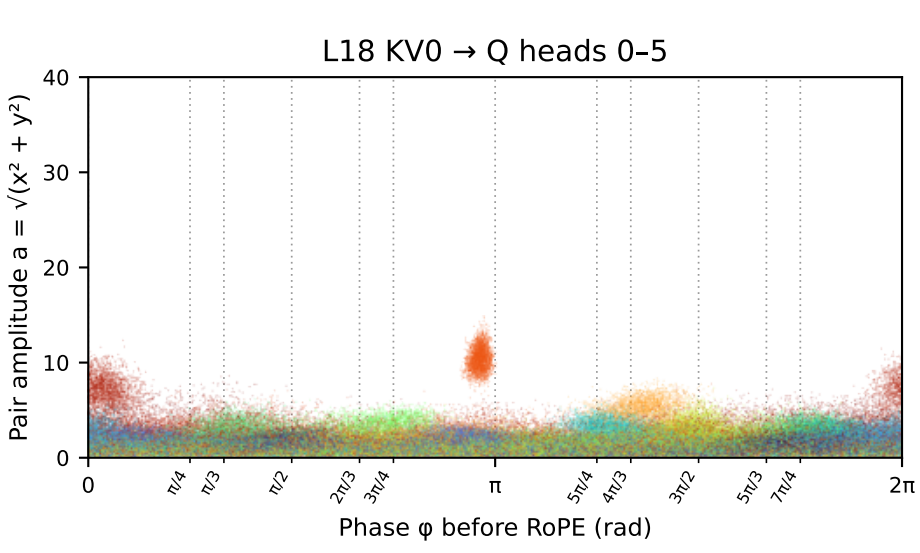
Qwen keys · layer 16–19 · amplitude axis 0–40

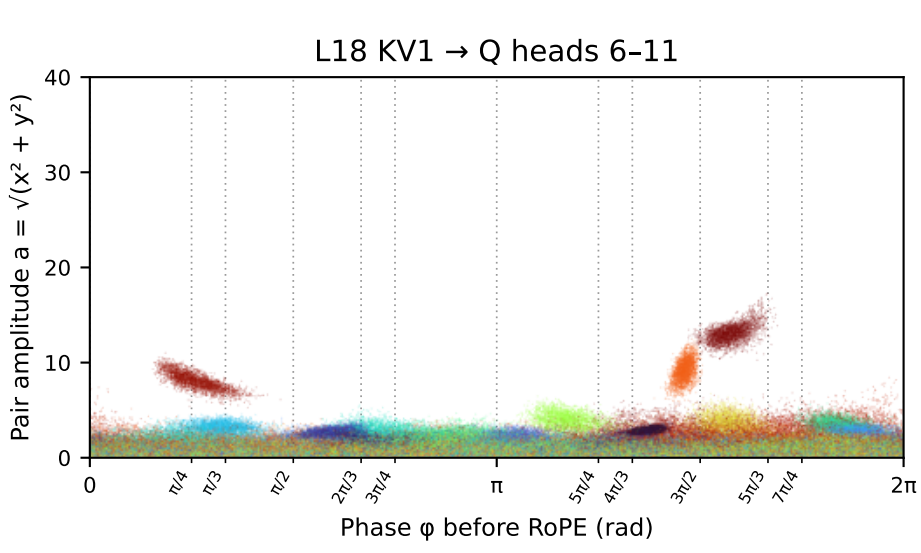


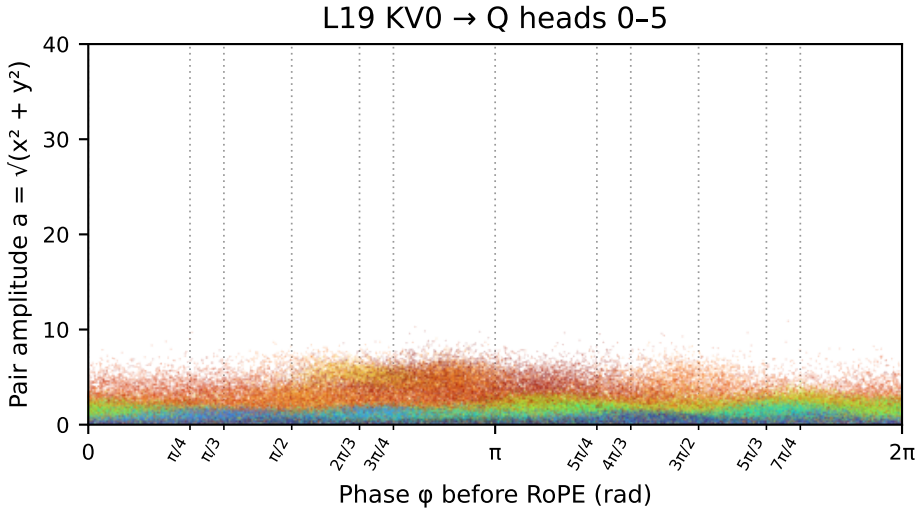


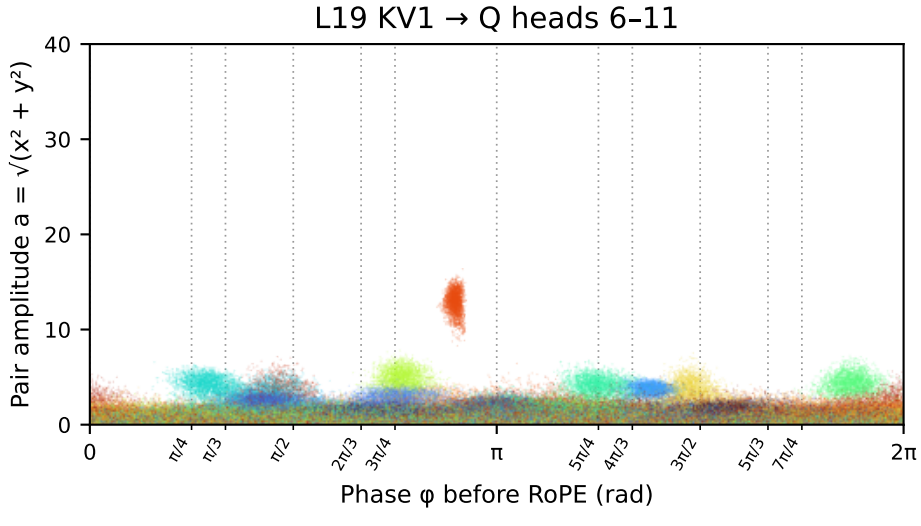


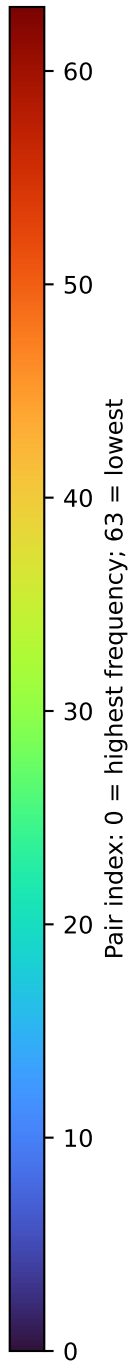






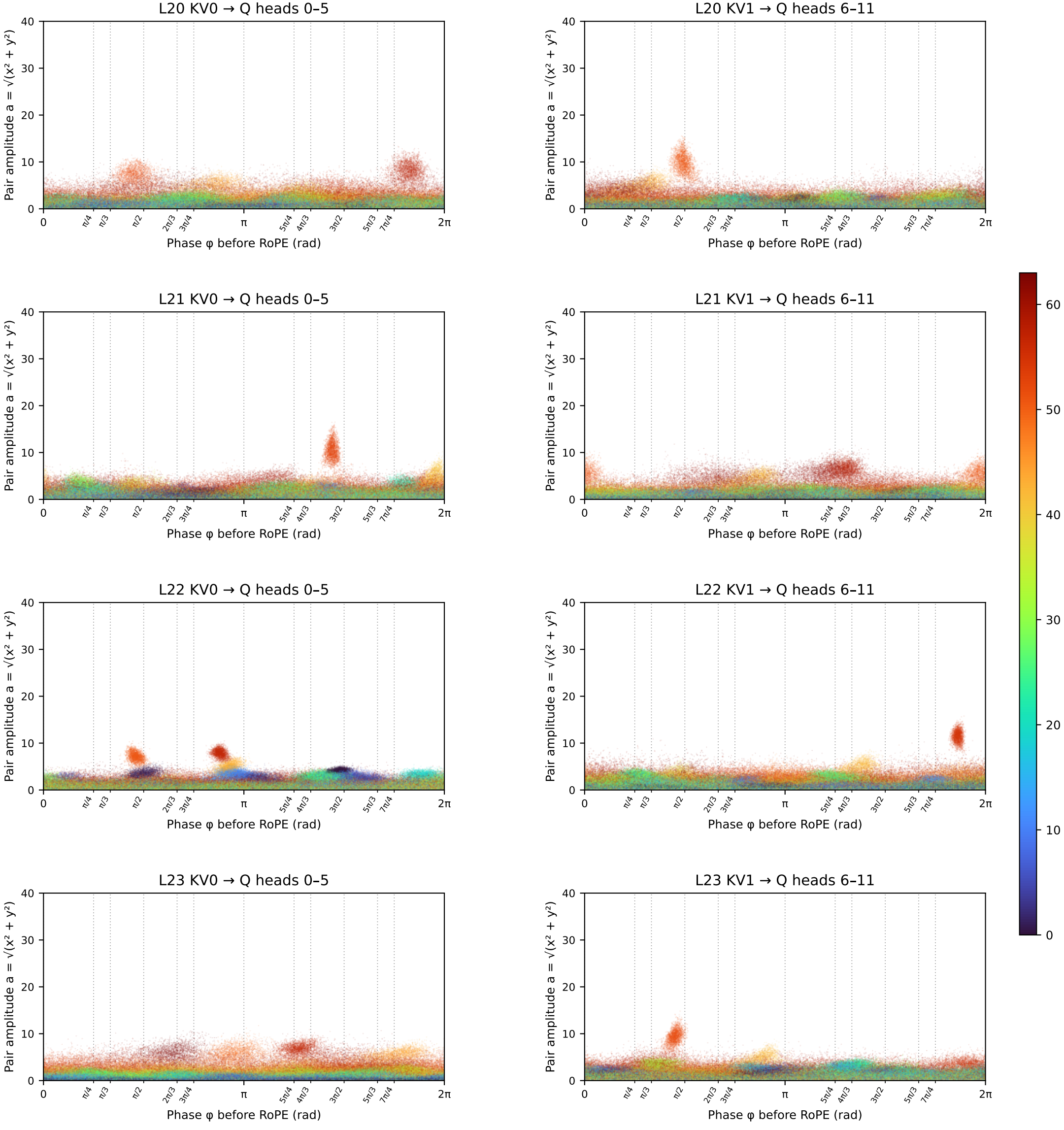






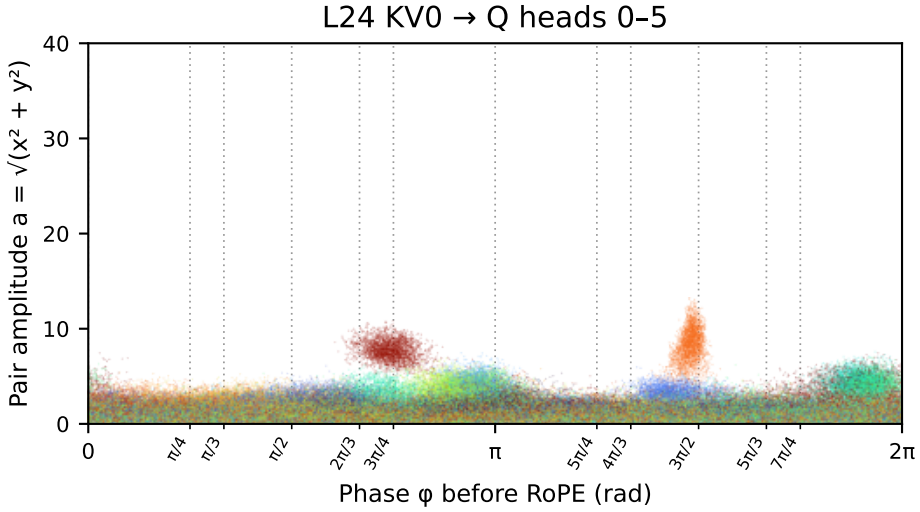
Every point = one token × one RoPE pair. 2,379 prose key tokens (chat-prefix keys excluded), 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Two shared key heads per layer: KV0 serves Q heads 0–5; KV1 serves Q heads 6–11.
Gray dotted guides mark multiples of π/4 and π/3.

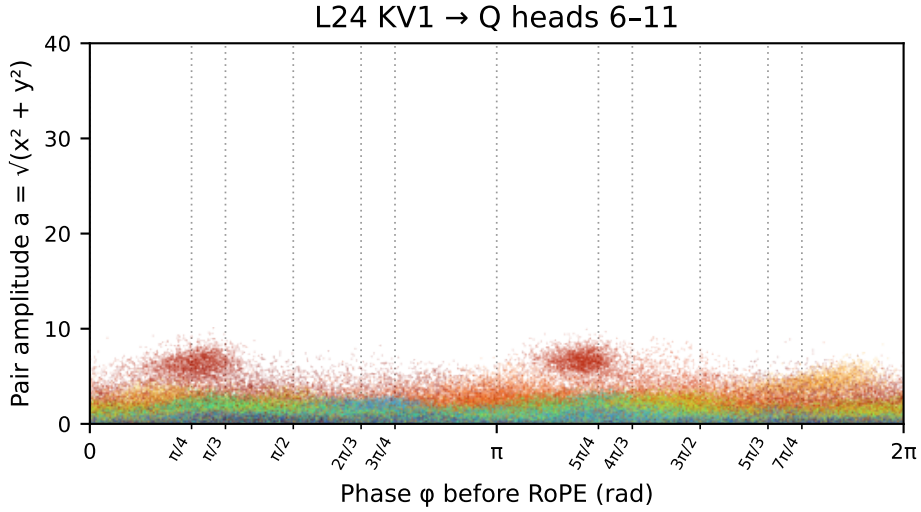
Qwen keys · layer 20–23 · amplitude axis 0–40

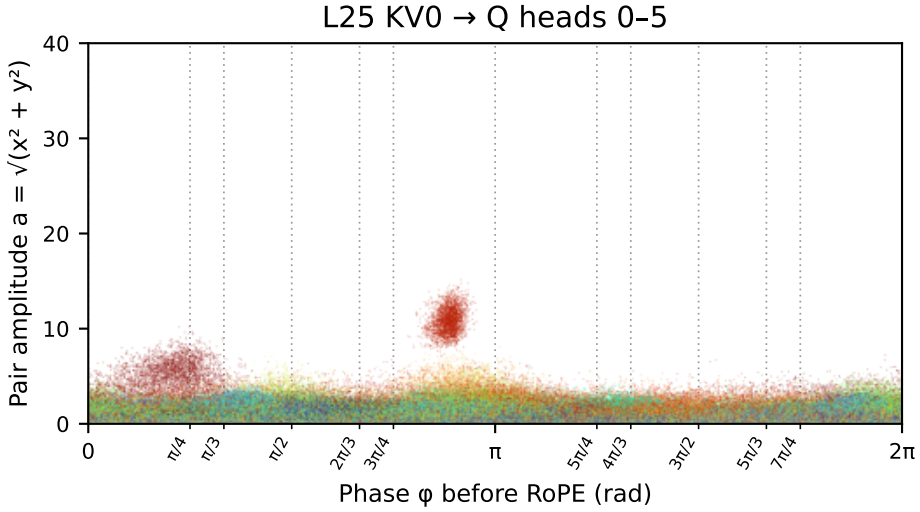


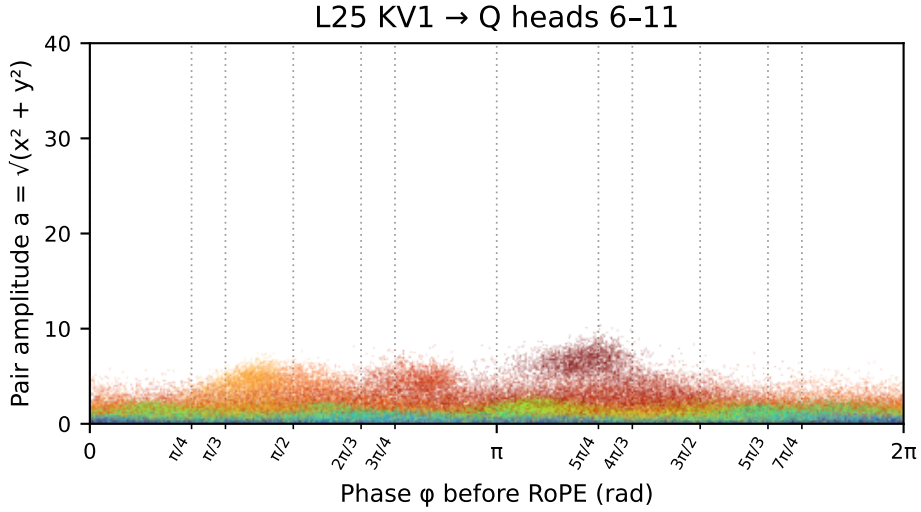
Every point = one token × one RoPE pair. 2,379 prose key tokens (chat-prefix keys excluded), 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Two shared key heads per layer: KV0 serves Q heads 0–5; KV1 serves Q heads 6–11.
Gray dotted guides mark multiples of π/4 and π/3.

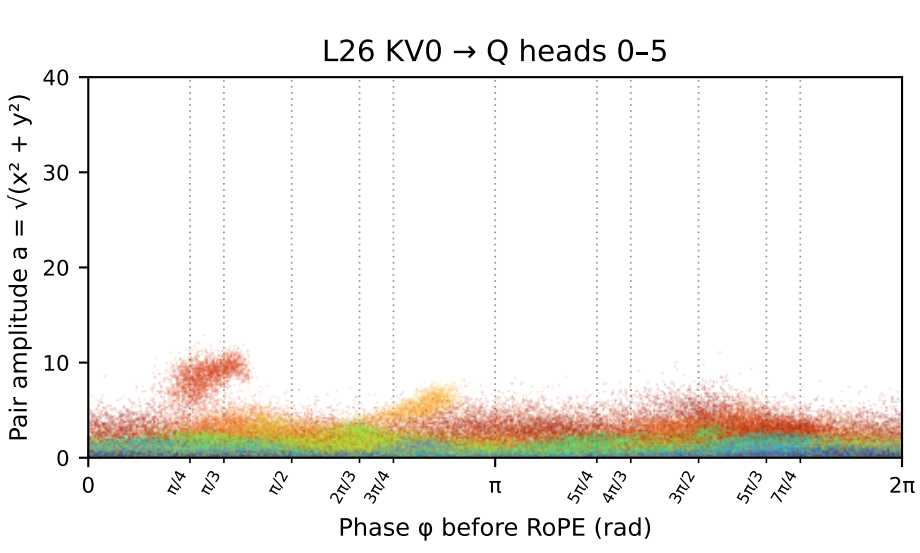
Qwen keys · layer 24–27 · amplitude axis 0–40

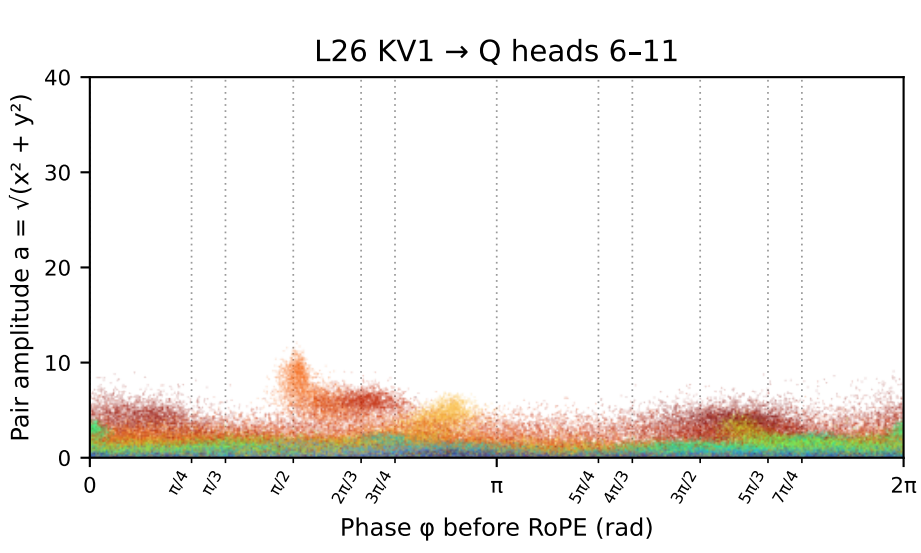


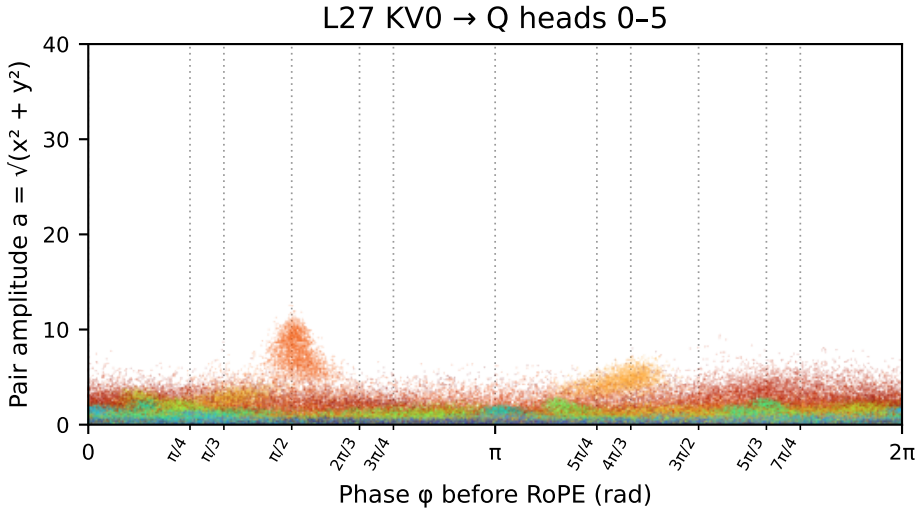


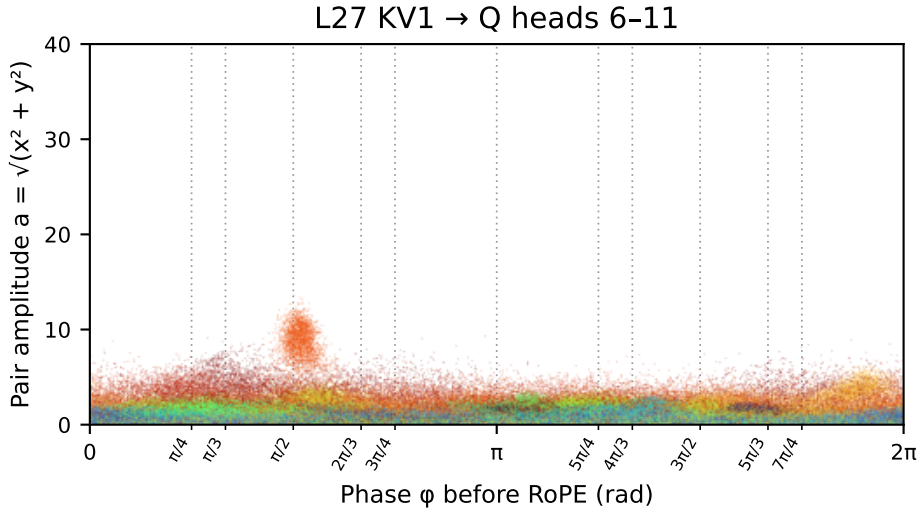


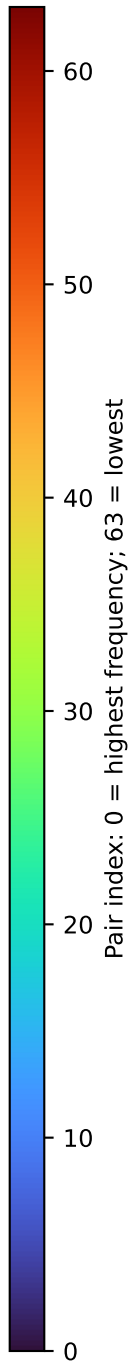












Every point = one token × one RoPE pair. 2,379 prose key tokens (chat-prefix keys excluded), 64 pairs per head.
Known positional rotation removed. Linear axes; points above amplitude 40 are clipped.
Two shared key heads per layer: KV0 serves Q heads 0–5; KV1 serves Q heads 6–11.
Gray dotted guides mark multiples of π/4 and π/3.